How do we implement gates? (Single qubit)



Atom (quantized) ⊕ classical field

Interaction Hamiltonian

$H = H_0 + H_I$

H_0: atom bit; H_I: interaction w/ field.



Two level system

$$H_0|\psi_1\rangle = E_1|\psi_1\rangle \; ; \quad H_0|\psi_2\rangle = E_2|\psi_2\rangle$$

$$\psi_n(r,t) = \psi_n(r)\, e^{-iE_n t/\hbar} \; ; \quad \frac{E_n}{\hbar} = \omega_n$$

Interaction Hamiltonian

$$H_I(t) = e\vec{r}\cdot\vec{E} \; , \quad \vec{E} = \frac{\vec{E}_0}{2} e^{i\omega t} + C.C.$$

→ leads to mixing the two states:

$$|\psi\rangle = \sum_n c_n e^{-i\omega_n t}|\psi_n\rangle$$

Plug in to $-i\hbar \frac{\partial |\psi\rangle}{\partial t} = H|\psi\rangle$

$$\sum (\hbar\omega_n c_n + i\hbar \dot{c}_n) e^{-i\omega_n t}|n\rangle = \sum_n (\hbar\omega_n + H_I) c_n e^{-i\omega_n t}|n\rangle$$

Multiply through by $\langle m| e^{i\omega_m t}$

$$i\dot{c}_n(t) = \frac{1}{\hbar} \sum_m \langle m|H_I|n\rangle e^{i\omega_{mn} t} c_m(t) \; ; \; \omega_{mn} = \omega_m - \omega_n$$

if non zero, states are coupled.

Back to two level system

$$i\dot{c}_1 = \Omega \cos(\omega t) c_2 e^{i\omega_0 t}$$
$$i\dot{c}_2 = \Omega^* \cos(\omega t) c_1 e^{-i\omega_0 t}$$

Define:

$\omega_0 = \omega_2 - \omega_1$

$\Omega = \langle 1| e\vec{r}\cdot\vec{E}_0|2\rangle/\hbar$

Rabi frequency.

$\Delta = \omega - \omega_0$ (detuning)

Solving coupled equations:

Expand cosine:

$$i\dot{c}_1 = \frac{\Omega c_2}{2}\left[e^{i(\omega-\omega_0)t} + e^{-i(\omega+\omega_0)t}\right]$$

(Δ; rapid term avg. to zero.) (rotating wave approx)

Define:

$$c_2(t) = \tilde{c}_2(t) e^{+i\Delta t}$$

Rewrite:

$$i\dot{c}_1 = \frac{\Omega}{2}\tilde{c}_2 \quad (1)$$

$$i\dot{c}_2 = \frac{\Omega^*}{2}c_1 + \Delta\tilde{c}_2 \quad (2)$$

Trial solution: $c_1(t) = e^{-iKt} \longrightarrow \tilde{c}_2(t) = \frac{2K}{\Omega}e^{-iKt}$

Get equation for K:

$$K^2 + \Delta K - \left|\frac{\Omega}{2}\right|^2 = 0$$

$$K_\pm = \frac{1}{2}\left(-\Delta \pm \sqrt{\Delta^2+\Omega^2}\right)$$

$$= \frac{1}{2}\left(-\Delta \pm \Omega'\right); \quad \Omega' = \sqrt{\Delta^2+\Omega^2}$$

generalized Rabi frequency.

Assume init. cond.

$$c_1(0) = 1 \; ; \; c_2(0) = 0.$$

$$c_1(t) = e^{i\Delta t/2}\left[\cos(\Omega' t/2) + \frac{i\Delta}{\Omega'}\sin(\Omega' t/2)\right]$$

$$c_2(t) = -\frac{i\Omega}{\Omega'}e^{i\Delta t/2}\sin(\Omega' t/2)$$

Probab. of finding system in state 2:

$$|c_2|^2 = \left|\frac{\Omega}{\Omega'}\right|^2 \sin^2(\Omega' t/2) \quad (\omega \neq \omega_0)$$

$$= \sin^2(\Omega t/2).$$

General case:

$$\vec{C}(t_0) = \begin{pmatrix} C_1(t_0) \\ C_2(t_0) \end{pmatrix} \; ; \quad \vec{C}(t) = U(t,t_0)\vec{C}(t_0)$$

For $t_0 = 0$:

$$U(t) = \begin{pmatrix} e^{i\Delta t/2}\left[\cos\left(\frac{\Omega' t}{2}\right) + \frac{i\Delta}{\Omega'}\sin\left(\frac{\Omega' t}{2}\right)\right] & -ie^{i\Delta t/2}\frac{\Omega^*}{\Omega'}\sin\left(\frac{\Omega' t}{2}\right) \\ -ie^{-i\Delta t/2}\frac{\Omega}{\Omega'}\sin\left(\frac{\Omega' t}{2}\right) & e^{-i\Delta t/2}\left[\cos\left(\frac{\Omega' t}{2}\right) - \frac{i\Delta}{\Omega'}\sin\left(\frac{\Omega' t}{2}\right)\right] \end{pmatrix}$$

On resonance: $\Delta = 0$

$$U(t) = \begin{pmatrix} \cos(\Omega t/2) & -i\sin(\Omega t/2) \\ -i\sin(\Omega t/2) & \cos(\Omega t/2) \end{pmatrix}$$

Additional control: phase of driving field.

$e^{i\Delta t/2} = e^{i(\omega - \omega_0)t/2}$ | could have an arbitrary phase ϕ_0 on driving field.

$\longrightarrow$ $e^{i\phi_0 t}$ applied to off diagonal elements.

Realising single qubit gates: ⟶ need right $\pi/2$

① Realising an X-gate: $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$t = \frac{\pi}{n}$ $\quad U\left(\frac{\pi}{n}\right) = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} = U_\pi$

π-pulse $\quad$ X-gate w/ global phase $e^{-i\pi/2}$

② **2π pulse:**

$$U_{2\pi} = U_\pi U_\pi = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}\begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} = -\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

note phase shift

③ **$\pi/2$ pulse:** ; $t = \pi/2|n|$

$$U_{\pi/2} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix}$$

④ Y-gate? (use phase shift of driving field: $\varphi = \pi/2$)

do π pulse

$$U_{\pi,y} = \begin{pmatrix} 0 & -e^{i\pi/2} i \\ -e^{i\pi/2} i & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$e^{i\pi/2} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \longrightarrow \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \longrightarrow \text{Y gate.}$$

⑤ Z-gates: → use composite pulses:

$$R_z(\theta) = e^{i\alpha} R_x(\beta) R_y(\gamma) R_x(\delta)$$

$$R_z(\pi) = R_x(3\pi/2) R_y(\pi) R_x(\pi/2)$$

$$R_z(\pi/2) = R_x(\pi/2) R_y(3\pi/2) R_x(3\pi/2)$$

Alternatively:

$$U(\delta) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\delta} \end{pmatrix}$$

Achieved by shining laser resonant (near) w/ upper state.